

Linux Interview Questions for DevOps

72 frequently asked questions — Basics to Advanced Troubleshooting

Basics & File System

1. Linux file system hierarchy (FHS) explain karo — /etc, /var, /usr, /opt kya hota hai?
2. Soft link vs Hard link mein difference?
3. File permissions kaise kaam karti hain (rwx, chmod, chown)?
4. chmod 755 ka matlab kya hota hai?
5. Inode kya hota hai Linux mein?
6. find aur locate command mein difference?
7. /etc/passwd aur /etc/shadow file mein kya hota hai?
8. Symbolic link kaise banate ho (ln -s)?

Process Management

9. Process kya hota hai aur PID kya hai?
10. ps, top, aur htop commands mein difference?
11. Zombie process kya hota hai aur usko kaise handle karte ho?
12. Foreground vs Background process — &, nohup, fg, bg ka use?
13. kill, kill -9, aur killall mein difference?
14. Load average kya hota hai (jo top mein dikhta hai)?
15. Cron job kaise schedule karte ho? Crontab syntax batao.

Networking

16. netstat ya ss command ka use kya hai?
17. Port kya hota hai aur kaise check karte ho konsa port open hai?
18. ping, traceroute, curl, wget mein difference aur use case?
19. DNS resolution kaise kaam karti hai Linux mein? /etc/hosts vs DNS server?
20. Firewall rules kaise manage karte ho (iptables/firewalld)?
21. SSH kaise kaam karta hai? Key-based authentication kaise setup karte ho?
22. scp aur rsync mein difference?

System Administration

23. Disk usage check karne ke commands (df -h, du -sh)?
24. Memory usage kaise check karte ho (free -m)?

- 25. Log files kahan store hote hain (/var/log) aur journalctl ka use?
- 26. System boot process explain karo (BIOS to Bootloader to Kernel to Init/systemd).
- 27. systemctl commands — service start/stop/enable/disable kaise karte ho?

DevOps-Practical Scenarios

- 28. Server ka CPU 100% pe chal raha hai, troubleshoot kaise karoge? (Step by step approach)
- 29. Disk space full ho gaya production server mein — kaise identify aur fix karoge?
- 30. Ek application crash ho rahi hai baar baar — logs se root cause kaise dhundhoge?

Shell Scripting & Text Processing

- 31. Shell script mein variable kaise declare karte ho, aur environment variable vs shell variable mein difference?
- 32. grep, awk, aur sed mein basic difference aur use case batao.
- 33. Ek shell script likho jo kisi directory ke saare .log files delete kare 7 din se purane.
- 34. \$1, \$2, \$@, \$# ka use shell script mein kya hota hai?
- 35. Exit status/exit code kya hota hai (\$)? Error handling script mein kaise karte ho?
- 36. cut, sort, uniq commands ka use kab karte ho (real example do)?
- 37. Pipe (|) aur redirection (>, >>, 2>) mein difference?
- 38. Cron job fail ho raha hai silently — debug kaise karoge?

Users, Permissions & Security

- 39. sudo vs su mein difference?
- 40. New user kaise create karte ho aur usko specific group mein add karte ho (useradd, usermod)?
- 41. SUID, SGID, aur Sticky bit kya hote hain?
- 42. SELinux ya AppArmor kya hote hain, inka role kya hai?
- 43. Password-less SSH login kaise setup karte ho servers ke beech?
- 44. Server hardening ke liye kaunse basic steps lete ho (security best practices)?

Storage, Package Management & Kernel

- 45. apt vs yum/dnf — package management mein difference?
- 46. LVM (Logical Volume Manager) kya hota hai aur kyun use karte hain?
- 47. Swap space kya hota hai aur kaise configure karte ho?
- 48. mount aur /etc/fstab ka use kya hota hai?
- 49. Kernel parameters kaise tune karte ho (sysctl, /proc filesystem)?

50. Kernel panic kya hota hai aur usse kaise troubleshoot karte ho?

Performance & Troubleshooting

51. vmstat aur iostat commands ka use kya hota hai?

52. High disk I/O wait ho raha hai server pe — kaise identify karoge?

53. strace aur lsof commands kab use karte ho troubleshooting mein?

54. Server pe “Too many open files” error aaye toh kya karoge? (ulimit concept)

55. Memory leak kaise detect karoge ek running process mein?

56. dmesg command kya batata hai aur kab check karte ho?

57. Network latency issue ho rahi hai do servers ke beech — kaise debug karoge?

Filesystem & Storage Deep-dive

58. Ext4 vs XFS filesystem mein difference?

59. RAID kya hota hai? RAID 0, 1, 5, 10 mein basic difference batao.

60. Disk full hai lekin df -h khaali dikha raha hai — ye kaise possible hai? (deleted-but-open-file scenario)

61. tar, gzip, aur zip mein difference aur use kab karte ho?

62. NFS kya hota hai aur kaise mount karte ho?

Automation & Config Management

63. Ansible playbook kya hoti hai, basic structure batao (Linux ke context mein).

64. Idempotency kya hoti hai automation scripts/tools mein?

65. screen aur tmux ka use kya hota hai?

66. Systemd unit file kaise likhte ho custom service ke liye?

67. Log rotation kaise setup karte ho (logrotate)?

Real-World Scenarios

68. Production server achanak unreachable ho gaya — step by step kaise approach karoge?

69. Ek cron job duplicate run ho raha hai (do baar chal raha hai) — kya reason ho sakta hai aur fix kaise karoge?

70. Server reboot hone ke baad ek zaroori service start nahi hui — kaise troubleshoot aur permanently fix karoge?

Quick Reference: chmod vs chown vs chgrp

chown (Change Owner) — file/directory ka owner AUR group dono change karta hai. Syntax: *chown user:group filename* | Recursive: *chown -R user:group /path/*

chgrp (Change Group) — sirf group ownership change karta hai, owner wahi rehta hai. Syntax: *chgrp groupname filename* | Recursive: *chgrp -R groupname /path/*

Command	Kya change karta hai
chown	Owner + Group dono
chgrp	Sirf Group
chmod	Permissions (rwx)

Tip: Shell scripting, troubleshooting scenarios, aur systemctl/cron/logs — ye teeno almost har DevOps Linux interview mein aate hain. In pe extra practice zaroor karo.